DRILL 28

NFPA 1001 Level 2 (JPR 5.5.1) / NFPA 1962

APPARATUS & EQUIPMENT MAINTENANCE

ANNUAL SERVICE HOSE TESTING & INSPECTION

⏱ **Time:** 60–75 min | 🚒 **Apparatus:** Pumper / Hose Testing Rig | 🦺 **PPE:** Helmet, Safety Glasses, Boots & Gloves

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Testing & Inspection Crew

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must know how to inspect fire hose for cuts, chemical degradation, coupling damage, and missing gaskets, connect hose to a test manifold or pumper discharge with a safety test gate valve, bleed all air from the lines, pressurize the layout to **NFPA 1962 service test pressure**, hold the pressure for **5 minutes**, and safely mark or tag out defective sections. A burst hose line inside a burning building is catastrophic.

📦 TOOLS & APPARATUS SETUP

- Pumper or specialized electric hose tester with test gate valve (valve with 1/4" bleeder hole to prevent pressure surges if a hose bursts).
- Sections of 1.75" (45mm) attack line and 4"/5" (100mm/125mm) LDH supply line.
- Test nozzles or threaded test caps with petcock bleeder valves.
- Spanner wrenches, grease pencil/paint marker, hose record log sheets.
- 100-ft tape measure.

⚠ SAFETY WATCH-OUTS & STOP RULE

⚠ **Catastrophic Burst Hazard:** During the 5-minute pressurized hold, **NO PERSONNEL MAY STRADDLE OR WALK ALONG THE HOSE LINES!** Stand a minimum of **15 feet to the side** (the "safe zone").

⚠ **Trapped Air Hazard:** Air compresses under pressure; water does not. Trapped air turns a bursting hose into an explosive whip. **ALL AIR MUST BE COMPLETELY BLED** before raising pressure!

🛑 **THE STOP RULE: Shout "STOP!" immediately if anyone approaches a pressurized line, or if a coupling begins to slip noticeably from the hose collar.**

PHASE 1: VISUAL INSPECTION & COUPLING CHECK (15 MINUTES) — SLOW DRY RUN

1. Lay hose straight on flat pavement.
2. **Jacket Inspection:** Check for mold/mildew rot, cuts, heavy abrasion, burns, or chemical stains (battery acid, motor oil).
3. **Coupling & Gasket Inspection:**
   - Swivel must rotate smoothly by hand without binding.
   - Inspect female swivel gasket: Must be supple, free of cracks, and seated flat in the groove.
   - Check lugs and threads for dents, burrs, or flattening.
   - Check Storz locks for spring tension and intact locking levers.
4. **Marking the Coupling Seam:** Draw a line with a grease pencil around the hose jacket directly where it meets the metal coupling ferrule. (If the coupling slips during pressure testing, this pencil line will move, revealing a failing connection!).

DRILL 28: PRACTICAL EVOLUTIONS & TAILGATE DEBRIEF

Apparatus & Equipment Maintenance • Practical Application

PHASE 2: LAYING OUT & BLEEDING AIR (20 MINUTES) — BUDDY COACHING

1 Layout Rules:

- Maximum length of any single test line connected to a discharge is **300 feet (90m)**.
- Keep lines straight without sharp kinks or bends.

2 Connecting & Bleeding:

- Connect hose to pumper test gate valve or hose tester manifold.
- Connect test cap or shutoff nozzle to the far end.
- Open the test gate valve and charge the layout with low hydrant/tank pressure (~45–50 PSI).
- Elevate the far end test cap/nozzle slightly and open the bleeder petcock until all trapped air escapes in a steady stream of solid water.
- Close the bleeder valve and verify the test line is completely full of water with zero air pockets.

PHASE 3: PRESSURIZATION, 5-MINUTE HOLD & LOG SIGN-OFF (20 MINUTES) — TIMED / STRESS

1 Service Test Pressures (NFPA 1962 Standard):

- **Attack Hose (1.75" / 2.5"):** Typically tested at **250 to 300 PSI** (or original stenciled service test pressure).
- **Supply Line / LDH (4" / 5"):** Typically tested at **200 PSI**.

2 Ramping Pressure:

- Raise pump pressure slowly at a rate not exceeding **25 PSI per second**.
- Once target service test pressure is reached, close the test gate valve.
- Start the timer: Hold pressure for **5 MINUTES**.
- Personnel monitor lines from the designated safe zone 15 feet away.

3 Post-Test Inspection:

- After 5 minutes, bleed pressure down slowly to zero.
- Inspect coupling pencil marks: If the line moved more than 1/8 inch (3mm) away from the coupling, or if water weeping occurred around the shank, the hose **FAILS**.
- If any leaks, pinholes, or coupling slippage occur: Cut the defective hose coupling off or spray-paint a red "X" and tag out of service.
- Record hose ID numbers, test date, test pressure, and pass/fail status in the department NFPA log.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Testing hose with trapped air:* Air acts as a loaded spring. A burst hose with trapped air will explode and throw shrapnel; a hose with 100% water simply splits with a harmless splash.
- ❌ *Walking over charged test lines:* High-pressure testing exposes latent weaknesses in fabric. Never step over or straddle lines under test!
- ❌ *Ramping pressure too fast:* Spiking pressure instantly damages internal rubber liners and blows out pump packing.

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"Did any pencil marks move on our older hose couplings?", "Why must hose be tested annually under NFPA standards?"***

✅ APPARATUS RESET CHECKLIST

- ☐ All passed hose sections stamped/logged with test year and stowed back on apparatus beds.
- ☐ Any failed hose tagged, marked with red tape, and placed in repair bin.
- ☐ Test caps and manifold dried and returned to maintenance cabinet.
- ☐ Hose test log signed and filed for Fire Marshal / FUS records.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS